

# Extension Methods for the Hybrid Movie Recommender

Candidate approaches to test beyond the current baseline (Θέμα 2)

**Dataset:** MovieLens 25M | **Current pipeline:** Global Mean, Popularity, Content-Based, User-kNN, Item-kNN, SVD, Weighted Hybrid, Stacked Hybrid

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## 1. Purpose of this report

The current implementation already exceeds the assignment's minimum (three models required; eight built, including two hybrid strategies). This report catalogues *optional, advanced extensions* — methods worth testing to (a) make the hybrid genuinely contribute rather than collapsing onto SVD, (b) fix the weak ranking results, and (c) demonstrate methodological depth.

Each method is rated on **Impact** (how much it helps the specific weaknesses of this pipeline), **Effort**, and **On-topic fit** for Θέμα 2.

The two known issues from the project walkthrough — the F1@K stable-sort tie-break bug and the missing Stacked Hybrid row — are corrections, not extensions, and are assumed fixed before any of the below is attempted.

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## 2. Diagnosis: why the current hybrid underperforms its potential

Two facts drive the entire extension strategy:

1. **The content signal is weak.** The Weighted Hybrid beats SVD by only  $\sim 0.0013$  RMSE because  $\alpha$  tuned to roughly 95% SVD — the content-based branch carries almost no usable signal. Root cause: content features come from free-text tags only, while the **tag genome (1,128 dense relevance scores per movie)** is loaded but never used.
2. **The models optimize the wrong objective for ranking.** SVD minimizes squared error, not item ranking, so it scores poorly on Precision/Recall/F1@K even though it has the best RMSE. This is the classic *RMSE-optimal  $\neq$  ranking-optimal* gap.

Every method below targets one or both of these.

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### 3. Tier A — Highest impact, on-topic, low-to-medium effort

#### A1. Use the tag genome as the content representation

**Impact: High · Effort: Low · Fit: Core**

Replace or augment the 256-dim TF-IDF text block with the 1,128-dim genome relevance vector (optionally reduced via TruncatedSVD to keep dimensionality manageable). This is the single best move: it turns the content-based branch from noise into a genuinely informative signal, which is the precise thing the assignment is testing (that a real hybrid beats each method alone).

- **How:** Pivot `genome-scores.csv` into a movie  $\times$  tag matrix; L2-normalize; concatenate with (or replace) the existing text block; refit the SVD compressor in `02_features.ipynb` so serving stays leak-free.
- **Expected benefit:** A measurably larger non-SVD weight in the Weighted Hybrid's  $\alpha$  sweep, and improved cold-item coverage. Literature on tag-driven content features (e.g. TAG-CF) reports double-digit NDCG@10 gains on MovieLens, concentrated among low-interaction users.
- **Why it matters for grading:** Directly strengthens the required hybrid-vs-baseline comparison.

#### A2. Ranking-optimized objective (BPR)

**Impact: High · Effort: Medium · Fit: Core**

Swap the squared-error SVD objective for Bayesian Personalized Ranking (BPR-MF), which optimizes the *order* of items rather than absolute ratings. This attacks the exact metric (F1@K) where the pipeline is weakest.

- **How:** Use an implicit-feedback library (e.g. `implicit`), or RecBole's BPR) on the binarized relevance  $\geq 4.0$  matrix; add it as a new base model and as an input to the stacked hybrid.
  - **Expected benefit:** Substantially better Precision/Recall/F1@K than rating-prediction SVD, at the cost of RMSE/MAE (BPR does not predict rating values).
  - **Note:** Report BPR on ranking metrics only; keep SVD for the rating-accuracy table.
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### 4. Tier B — Advanced models (strong "extension" candidates)

#### B1. LightGCN (graph-based collaborative filtering) — headline extension

**Impact: Very High (ranking) · Effort: Medium-High · Fit: Extension**

A graph neural network that learns user/item embeddings by message-passing over the user-item interaction graph, trained with a ranking (BPR) loss. It is the strongest single modeling upgrade for the ranking metrics and is especially well-motivated here because published benchmarks use the *same* dataset.

- **Reference benchmark (MovieLens 25M):** LightGCN reached the lowest RMSE ( $\sim 0.809$ ) and the highest NDCG@10 ( $\sim 0.435$ ) in a 2025 comparative study, beating SVD++ by  $\sim 10\%$  on ranking. A tuned KNN baseline on the same data reached NDCG@10  $\approx 0.405$  / F1  $\approx 0.231$ , versus LightGCN's NDCG@10  $\approx 0.435$  / F1  $\approx 0.255$ .
- **vs predecessor:** LightGCN improves on NGCF by  $\sim 16\%$  on recall and NDCG by removing feature transformation and non-linear activation.
- **How:** RecBole or the Microsoft `recommenders` library both ship LightGCN with MovieLens loaders; build the bipartite adjacency from the binarized interactions, 2-4 propagation layers.
- **Thematic note:** Graph methods lean toward  $\Theta_{\mu} 1$  (link prediction), but as an *extension comparison* in  $\Theta_{\mu} 2$  they are legitimate — and they fit the repository's knowledge-graph theme.

## B2. Neural CF bases (NCF / DeepFM / Mult-VAE)

**Impact: Medium-High • Effort: Medium • Fit: Extension**

Add one or more neural base learners. DeepFM and LightGCN consistently outperform traditional methods on top-k ranking in the MovieLens benchmarks; Mult-VAE is a strong autoencoder-based CF model. These slot naturally into the existing stacked-hybrid meta-learner as extra base predictions.

- **How:** RecBole provides all three with consistent evaluation; train on the existing split, feed out-of-fold predictions into the Ridge meta-model.

## B3. Sequential recommenders (SASRec / BERT4Rec)

**Impact: Medium-High • Effort: Medium-High • Fit: Extension (great fit)**

Because the pipeline already uses a **temporal per-user split**, a sequential model that predicts the next item from the ordered rating history exploits signal the current static models discard. This is a clean, defensible advanced method that respects the existing protocol exactly.

- **How:** SASRec (self-attention) or BERT4Rec (masked-item modeling) via RecBole; evaluate next-item ranking against the same sampled/full negatives.
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## 5. Tier C — Frontier / standout demo

### C1. LLM-as-reranker

**Impact: Medium (accuracy) · High (explainability/demo) · Effort: Medium · Fit: Extension**

Take the top-N candidates from SVD or LightGCN and have an LLM re-order them using the user's rated history expressed in natural language. Cheap, high-impact for the presentation, and produces human-readable justifications per recommendation.

- **How:** For a sampled user, prompt the LLM with "liked" titles + candidate list, ask for a re-ranked top-K with one-line rationales; evaluate the re-ranked list with the same metrics.

### C2. Conversational / LLM-hybrid recommender

**Impact: Medium · Effort: Medium-High · Fit: Extension**

An LLM interprets free-text user preferences and its suggestions are merged with the content-based candidate list — keeping grounded, deterministic similarity while adding nuanced preference understanding. This is the 2025–2026 frontier (generative / LLM-based recommendation) and makes a memorable app demo for requirement E.

- **Caveat:** Treat the LLM as a re-ranker or candidate enricher over the trained models, not as a standalone recommender, so the quantitative comparison stays honest.

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## 6. Evaluation upgrades (test alongside any method above)

These improve the *results* section (20% of the grade) regardless of which models are added.

| Upgrade                                                                    | Why                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Add <b>NDCG@K</b> and <b>AUC</b> beside <b>F1@K</b>                        | Standard ranking metrics, far less brittle to ties than the current <b>F1@K</b> .                                                                                                                                                                                                                                           |
| Run at least one <b>full-catalogue</b> ranking eval (no sampled negatives) | Sampled metrics with only 100 negatives are known to be inconsistent with exact metrics — they do not preserve "A beats B" even in expectation, and in the small-sample limit all metrics collapse toward AUC (Krichene & Rendle, KDD 2020). A full-catalogue pass on a user subsample makes the ranking claims defensible. |
| Apply <b>sampled-metric corrections</b> if full eval is too costly         | The same line of work proposes corrections that reduce sampled-metric bias.                                                                                                                                                                                                                                                 |
| Keep <b>rating</b> vs <b>ranking</b> tables separate                       | RMSE/MAE for rating-prediction models; NDCG/Recall/F1/AUC for ranking-optimized models (BPR, LightGCN, sequential).                                                                                                                                                                                                         |

## 7. Recommended priority plan

A realistic, maximum-value sequence:

1. **(Corrections — mandatory)** Fix the **F1@K** tie-break (shuffle candidates), re-run notebook 03 so all 8 models appear, and demonstrate the app on a few sampled dataset users (requirement E).
2. **(A1) Tag genome** into the content features — makes the hybrid real.
3. **(A2) BPR** base model — directly lifts the ranking metrics.
4. **(B1) LightGCN** — headline advanced comparison, benchmarked on the same dataset.
5. **(Evaluation)** Add NDCG + AUC and one full-catalogue ranking pass.
6. **(C1) LLM re-ranker** — optional "wow" extension for the presentation.

Items 1–3 alone move the submission from "good engineering, questionable ranking results" to "honest, complete, and competitive." Items 4–6 are what distinguish a strong project from a standout one.

## 8. References

1. Comparative study of collaborative filtering models on MovieLens 100K/1M/25M (LightGCN, SVD++, NCF, DeepFM, KNN variants). *Scientific Reports*, 2025.  
<https://www.nature.com/articles/s41598-025-15096-4>
2. He et al. *LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation*. SIGIR 2020. <https://arxiv.org/abs/2002.02126>
3. Krichene & Rendle. *On Sampled Metrics for Item Recommendation*. KDD 2020.  
<https://dl.acm.org/doi/10.1145/3394486.3403226>
4. *Towards Reliable Item Sampling for Recommendation Evaluation*. 2022.  
<https://arxiv.org/abs/2211.15743>
5. *A Survey on Generative Recommendation: Data, Model, and Tasks*. 2025.  
<https://arxiv.org/abs/2510.27157>
6. TAG-CF: tag-driven content features for collaborative filtering (NDCG@10 gains on MovieLens, esp. low-degree users). OpenReview. <https://openreview.net/forum?id=xwPket4FXm>
7. Microsoft (recommenders) MovieLens benchmark notebook (LightGCN, NCF, etc.).  
[https://github.com/recommenders-team/recommenders/blob/main/examples/06\\_benchmarks/movielens.ipynb](https://github.com/recommenders-team/recommenders/blob/main/examples/06_benchmarks/movielens.ipynb)

*Benchmark figures are directional comparisons from published studies; exact numbers depend on preprocessing, split, and negative-sampling choices, so reproduce them on your own pipeline rather than citing them as guarantees.*